

Introduction to Computer Vision

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● COURSE OBJECTIVES:

- To review image processing techniques for computer vision
- To illustrate shape and region analysis
- To describe Hough Transform and its applications to detect lines, circles, ellipses
- To discuss three-dimensional image analysis techniques
- To discuss motion analysis
- To explore some applications of computer vision algorithms

● COURSE OUTCOMES:

After successful completion of this course, student will be able to

- 1 **CO1:** Describe different image representation, their mathematical representation and data structures used
- 2 **CO2:** Classify different Boundary Detection algorithm
- 3 **CO3:** Create a 3D object from given set of images
- 4 **CO4:** Detect a moving object in video using the concept of motion analysis
- 5 **CO5:** Recognize the object using the concept of computer vision

- **Introduction:** Image Formation - Image Model, Imaging Devices
- **Early Processing:** Recovering Intrinsic Structure, Filtering Image, Finding Local Edges, Range Information from Geometry, Surface Orientation, Optical Flow, Resolution Pyramids.
- **Boundary Detection:** Searching Near and Approximate Location, Hough Method for Curve Detection, Edge Following As Graph Searching, Edge Following As Dynamic Programming, Contour Following.
- **Region Growing:** Regions, Local Technique, Blob Coloring, Global Techniques, Splitting And Merging.
- **Texture:** Structural Models, Texture As A Pattern Recognition Problem, Texture Gradients.

- **Motion:** Motion Understanding, Optical Flow, Image Sequences.
- **Representation of 2-D Geometrical Structure:** Boundary Representation, Region Representation, Simple Shape Properties, Representation Of 3-D Structures, Solids And Their Representation, Surface Representation, Generalized Cylinder Representation, Volumetric Representation, Understanding Line Drawings.
- **Knowledge Representation and Use:** Knowledge Base Models And Processes, Se-Mantic Nets, Control Issues in Vision Systems.
- **Matching:** Aspects, Graph Theoretic Algorithms, Implementation, Matching In Practice
- **Inference:** First Order Predicate Calculus, Computer Reasoning, Production Systems, Scene Labeling, Active Knowledge.

- Ballard and Brown, “Computer Vision”, Prentice Hall Publication.
<https://homepages.inf.ed.ac.uk/rbf/BOOKS/BANDB/bandb.htm>
- Jain, Kasturi and Schunck, “Machine Vision”, McGraw-Hill Int’l Editions.
<https://cse.usf.edu/~r1k/MachineVisionBook/MachineVision.pdf>
- Forsyth and Ponce, “Computer Vision: A Modern Approach”.
- Gonzalez and Woods, “Digital Image Processing”, Addison Wesley, 1992.
- Anil K Jain, “Fundamentals of Digital Image Processing”, Prentice Hall of India, 1995.

Online Resources

- CV Online.

Link: <https://homepages.inf.ed.ac.uk/rbf/CVonline/>

- The Computer Vision Homepage.

Link: <http://www.cs.cmu.edu/~cil/vision.html>

- Computer Vision - NPTEL.

Link: https://onlinecourses.nptel.ac.in/noc19_cs58/preview

- Computer Vision.

Link: <https://www.youtube.com/@patternanalysismachineinte4095>

- **Computer vision** is an interdisciplinary field that deals with how computers can be made to gain **high-level understanding** from digital images or videos.
- From the perspective of engineering, it seeks to automate tasks that the **human visual system** can do.

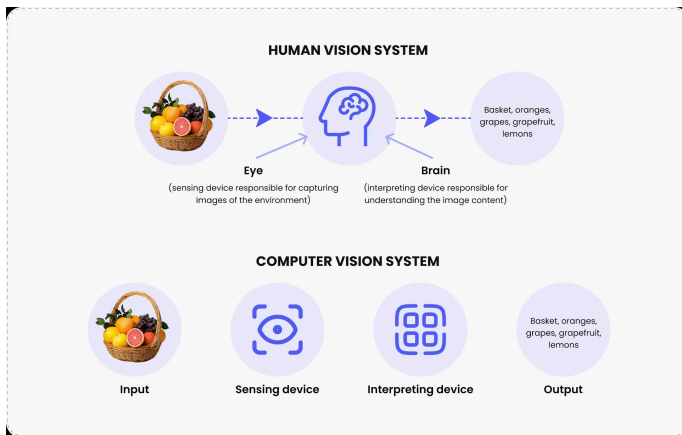


Figure 1: Human Vision and Computer Vision ¹

¹Source: <https://maddevs.io/blog/computer-vision-algorithms-you-should-know/>

- **Computer vision** is concerned with the
 - **automatic extraction**
 - **analysis**
 - **understanding**of useful information from a **single image** or a **sequence of images**.
- It involves the **development** of a theoretical and algorithmic basis to achieve **automatic visual understanding**.
- As a scientific discipline, computer vision is concerned with the theory behind artificial systems that **extract information from images**.

- The **image data** can take many forms, such as
 - video sequences
 - views from multiple cameras or
 - multi-dimensional data from a medical scanner.
- As a technological discipline, **computer vision** seeks to apply its theories and models for the construction of computer vision systems.
- **Machine vision** refers to a systems engineering discipline, especially in the context of **factory automation**.
- In more recent times, the terms computer vision and machine vision have converged to a greater degree.

- **Computer vision** tasks include methods for **acquiring, processing, analyzing, and understanding** digital images, and **extraction** of high-dimensional data from the real world in order to produce **numerical** or **symbolic information**, e.g. in the form of decisions.
- **Understanding** in this context signifies the transformation of visual images into **descriptions** of the world that **make sense** to thought processes and can elicit appropriate action.

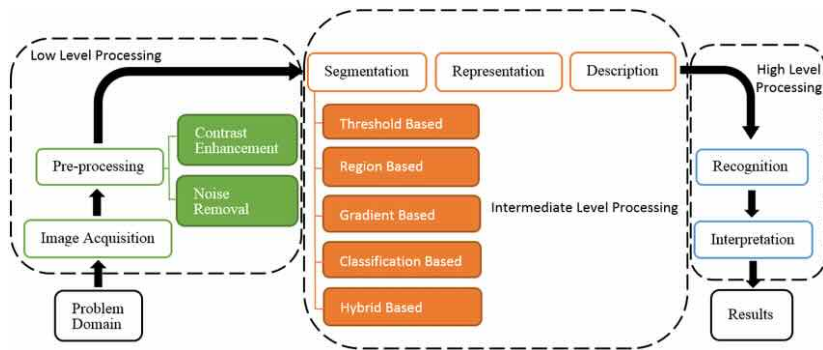


Figure 2: Steps of a Computer Vision System

Typical Tasks

Given a query image, try to find visually similar images from an image database

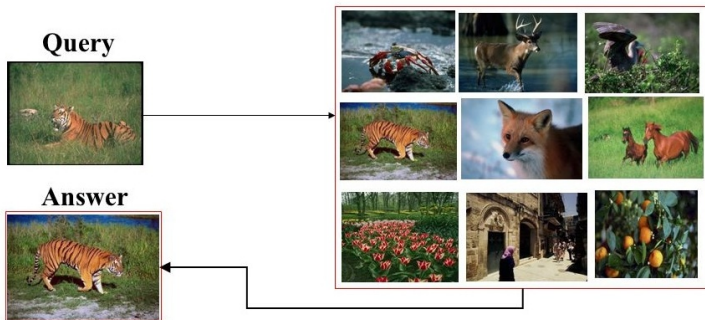


Figure 3: Content Based Image Retrieval ²

²Source: <https://slideplayer.com/slide/3415344/12/images/6/Content-based+Image+Retrieval.jpg>

Typical Tasks

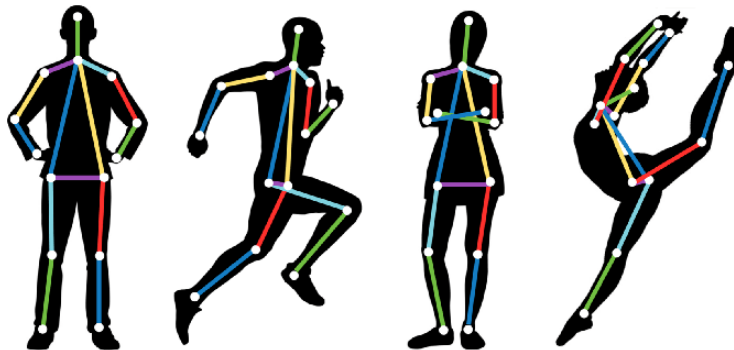


Figure 4: Pose Estimation ³

³Source: <https://kemtai.com/wp-content/uploads/2022/12/65603fig.png>

Typical Tasks

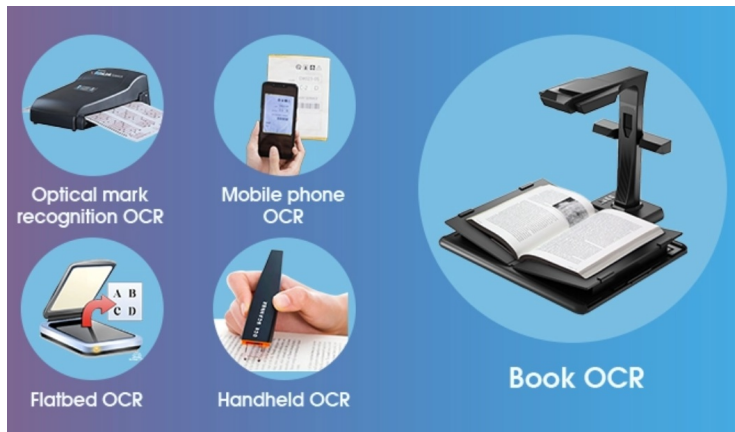


Figure 5: Optical Character Recognition ⁴

⁴Source: <https://macgence.com/blog/what-is-optical-character-recognition-ocr-and-how-does-it-work/>

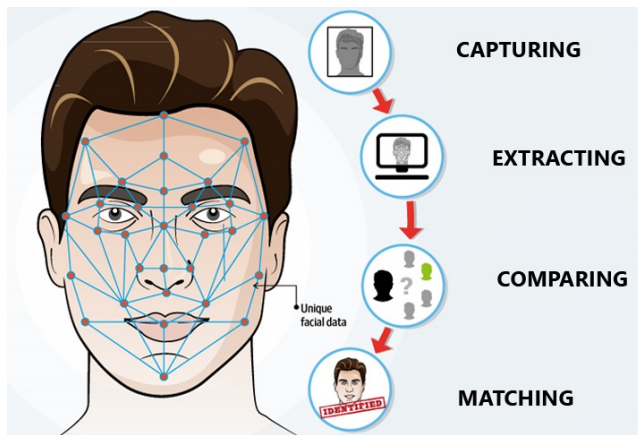


Figure 6: Face Recognition ⁵

⁵ Source: <https://www.starlinkindia.com/blog/biometrics-face-recognition/>

Typical Tasks



Figure 7: Emotion Recognition ⁶

⁶Source: <https://www.nature.com/articles/s41598-023-35446-4/figures/1>

Typical Tasks

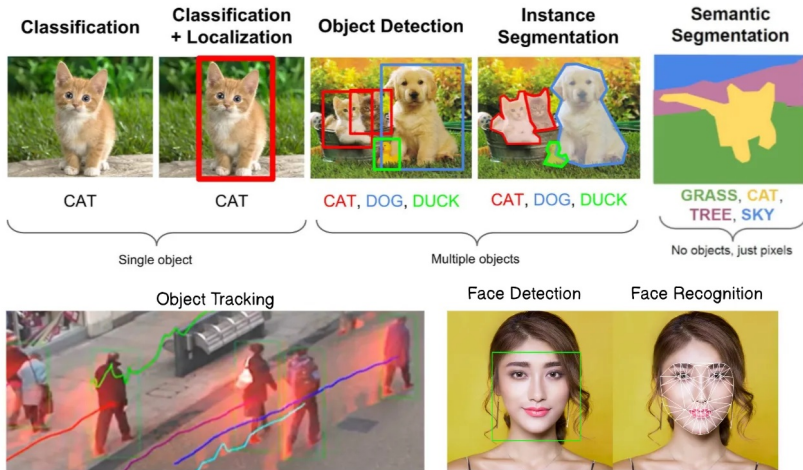


Figure 8: Different Problems in Computer Vision ⁷

⁷ Source: <https://medium.com/@fraidoonmarzai99/introduction-to-computer-vision-cv-in-depth-81a9ffc1b9f1>

- **Applications** of computer vision include
 - Scene Reconstruction
 - Object Detection
 - Event Detection
 - Activity Recognition
 - Video Tracking
 - Object Recognition
 - Learning
 - Indexing
 - Motion Estimation
 - Visual Servoing
 - 3D Scene Modeling
 - Image Restoration

- **Automatic inspection**, e.g., in manufacturing applications
- **Assisting humans** in identification tasks, e.g., a species identification system
- **Controlling processes**, e.g., an industrial robot
- **Detecting events**, e.g., for visual surveillance or people counting like in the restaurant industry
- **Interaction**, e.g., as the input to a device for computer-human interaction
- **Monitoring** agricultural crops, e.g. an open-source vision transformers model that helps farmers automatically detect strawberry diseases

- **Modeling objects or environments**, e.g., medical image analysis or topographical modeling
- **Navigation**, e.g., by an autonomous vehicle or mobile robot
- **Organizing information**, e.g., for indexing databases of images and image sequences
- **Tracking** surfaces or planes in 3D coordinates for allowing Augmented Reality experiences
- **Analyzing** the condition of facilities in industry or construction
- **Automatic real-time lip-reading** for devices and apps to assist people with disabilities

Image Analysis Tasks

<i>Domain</i>	<i>Objects</i>	<i>Modality</i>	<i>Tasks</i>	<i>Knowledge Sources</i>
Robotics	Three-dimensional outdoor scenes	Light	Identify or describe objects in scene	Models of objects
	indoor scenes	X-rays	Industrial tasks	Models of the reflection of light from objects
	Mechanical parts	Light		
		Structured light		
Aerial images	Terrain	Light	Improved images	Maps
	Buildings, etc.	Infrared	Resource analyses	Geometrical models of shapes
		Radar	Weather prediction	Models of image formation
			Spying	
			Missile guidance	
			Tactical analysis	
Astronomy	Stars	Light	Chemical composition	Geometrical models of shapes
	Planets		Improved images	
Medical Macro	Body organs	X-rays	Diagnosis of abnormalities	Anatomical models
		Ultrasound		Models of image formation
		Isotopes	Operative and treatment planning	
		Heat		
		Electronmicroscopy	Pathology, cytology	
Micro	Cells	Light	Karyotyping	Models of shape
	Protein chains			
	Chromosomes			
Chemistry	Molecules	Electron densities	Analysis of molecular compositions	Chemical models
			Structured models	
Neuroanatomy	Neurons	Light	Determination of spatial orientation	Neural connectivity
		Electronmicroscopy		
Physics	Particle tracks	Light	Find new particles	Atomic physics
			Identify tracks	

Figure 9: Examples of Image Analysis Tasks ⁸

⁸ Ref: Ballard and Brown, "Computer Vision", Table 1.1, Page 11.

Imaging Devices

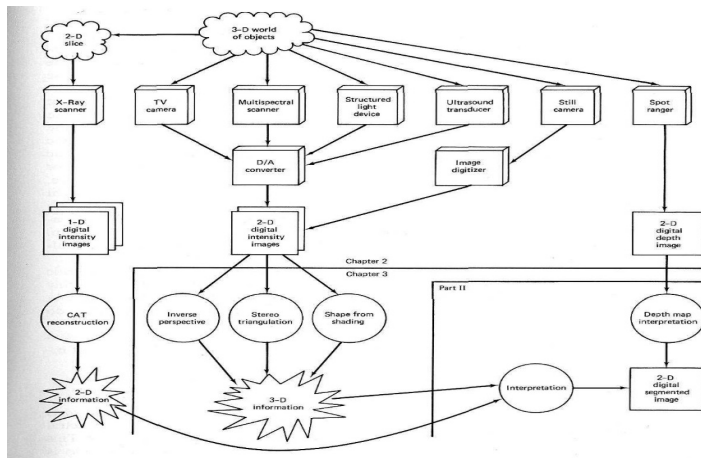


Figure 10: Imaging devices, information structures, and processes. ⁹

⁹ Ref: Ballard and Brown, "Computer Vision", Fig 2.16, Page 43.

Image Formation

- Image formation occurs when a sensor registers radiation that has interacted with physical objects.

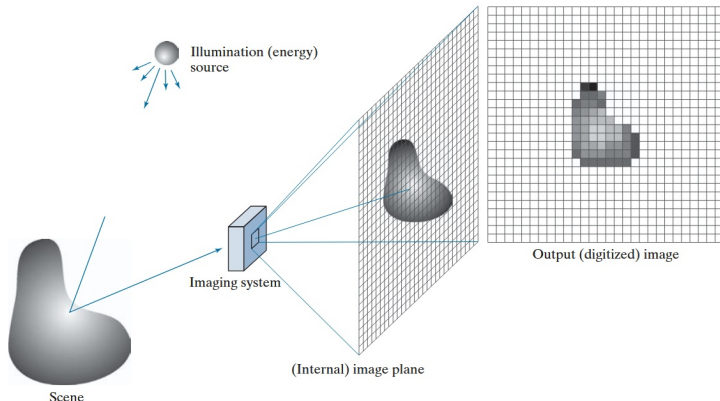


Figure 11: An example of digital image acquisition

- The **mathematical model** of imaging has several different components
 - An *image function* is the fundamental abstraction of an image.
 - A *geometrical model* describes how three dimensions are projected into two.
 - A *radiometrical model* shows how the imaging geometry, light sources, and reflectance properties of objects affect the light measurement at the sensor.
 - A *spatial frequency model* describes how spatial variations of the image may be characterized in a transform domain.
 - A *color model* describes how different spectral measurements are related to image colors.
 - A *digitizing model* describes the process of obtaining discrete samples.

- **Image Functions**

- An *image function* is a mathematical representation of an image.
- Generally, an image function is a vector-valued function of a small number of arguments.

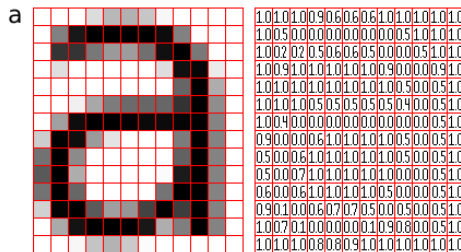


Figure 12: Image represented as a 2-D array ¹⁰

¹⁰ https://pippin.gimp.org/image-processing/chap_dir.html

• Image Functions

- Different image functions may be used to represent the same image, based on which of its **characteristics** are important.
- For instance, a camera produces an image on **black and white** film which is usually thought of as a real-valued function (whose value could be the density of the **photographic negative**) of two real-valued arguments, one for each of two spatial dimensions.
- However, at a very small scale (the order of the film grain) the negative basically has only two densities, **opaque** and **transparent**.

• Image Functions

- Most images are presented by functions of two spatial variables $f(x) = f(x, y)$, where $f(x, y)$ is the brightness of the gray level of the image at a spatial coordinate (x, y) .
- A *multi spectral* image f is a vector-valued function with components $(f_1, \dots, f_n)$.
- One special multi spectral image is a *color image* in which, for example, the components measure the brightness values of each of three wavelengths, that is,

$$f(x) = \left\{ f_{\text{red}}(x), f_{\text{blue}}(x), f_{\text{green}}(x) \right\} \quad (1)$$

- **Digital Image**

- A special case of the image function is the *digital (discrete)* image function, where the arguments to and value of the function are all integers.

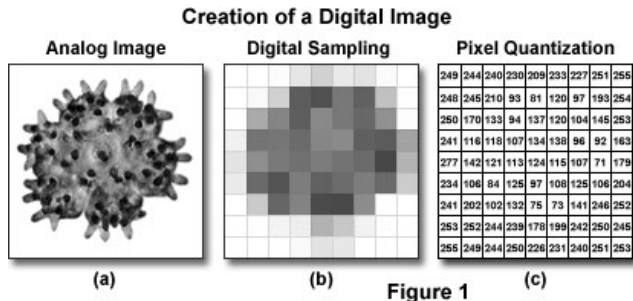


Figure 13: Creation of a Digital Image ¹¹

¹¹<https://hamamatsu.magnet.fsu.edu/articles/digitalimagebasics.html>

- **Pixel**

- In digital imaging, a **pixel** (a.k.a. **picture element**) is the smallest addressable element in a raster image, or the smallest addressable element in a dot matrix display device.
- In most digital display devices, pixels are the **smallest element** that can be manipulated through software.
- Each pixel is a sample of an original image; more samples typically provide more accurate representations of the original.

- **Pixel**

- The intensity of each pixel is variable.
- In **color** imaging systems, a color is typically represented by three or four component intensities such as **red, green, blue**, or **cyan, magenta, yellow, black**.
- Pixels can be used as a unit of measure such as: 2400 pixels per inch, 640 pixels per line, or spaced 10 pixels apart.

- **Pixel**

- The measures “**dots per inch**” (dpi) and “**pixels per inch**” (ppi) are sometimes used interchangeably, but have distinct meanings, especially for printer devices, where dpi is a measure of the printer’s density of dot (e.g. ink droplet) placement.
- For example, a high-quality photographic image may be printed with 600 ppi on a 1200 dpi inkjet printer.
- **Pixel counts** can be expressed as a single number,
 - as in a “three-megapixel” digital camera, which has a nominal three million pixels, or
 - as a pair of numbers, as in a “640 by 480 display”, which has 640 pixels from side to side and 480 from top to bottom and therefore has a total number of $640 \times 480 = 307,200$ pixels, or 0.3 megapixels.

Image Model

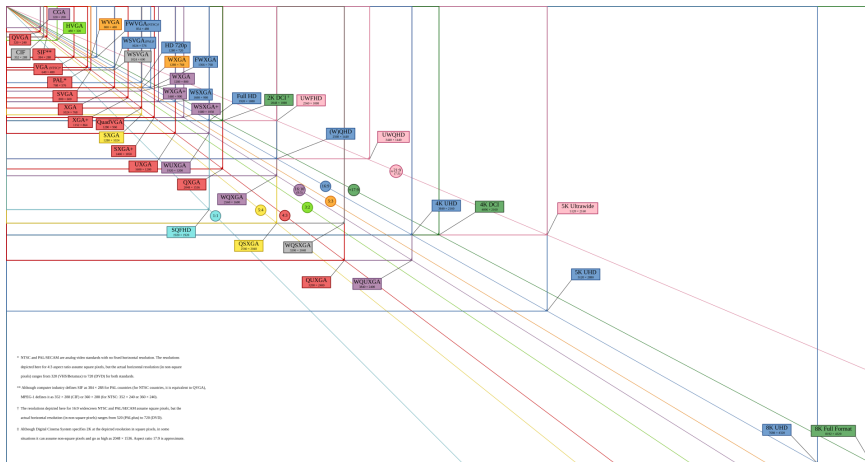


Figure 14: Image Resolutions¹²

¹²Source: https://en.wikipedia.org/wiki/Display_resolution

Name	Resolution (in Pixel)
540p / qHD	960×540
720p / HD	1280×720
1080p / Full HD (FHD)	1920×1080
2K	2048×1080
1440p / QHD / WQHD	2560×1440
2160p / UHD	3840×2160
4K	4096×2160
5K	5120×2880
8K / 8K UHD	7680×4320

Table 1: Common Display Resolutions ¹³

¹³Source: <https://www.whatismyscreenresolution.org/display-resolution-explained/>



Figure 15: Aspect Ratio ¹⁴

¹⁴ Source: <https://amadine.com/useful-articles/aspect-ratio>